

the line. We obtain a momentum space Feynman diagram with particles in states α and β on the mass shell plus r external lines of type $\ell, m \dots$ carrying momenta $p, p' \dots$ from the variational derivative

$$\left[\frac{\delta^r S_{\beta\alpha}[\epsilon]}{\delta\epsilon_\ell(x) \delta\epsilon_m(y) \dots} \right]_{\epsilon=0}$$

by stripping away the propagators on each of the off-shell lines and then taking the appropriate Fourier transforms and multiplying with appropriate coefficient functions u_ℓ, u_ℓ^* , etc. and a factor $(-i)^r$.

It is very useful for a number of purposes to recognize that there is a simple relation between the sum of contributions from all perturbation theory diagrams for any off-shell amplitude and a matrix element, between eigenstates of the full Hamiltonian, of a time-ordered product of corresponding operators in the Heisenberg picture. This relation is provided by a theorem,³ which states that to all orders of perturbation theory*

$$\left[\frac{\delta^r S_{\beta\alpha}[\epsilon]}{\delta\epsilon_a(x) \delta\epsilon_b(y) \dots} \right]_{\epsilon=0} = \left(\Psi_\beta^-, T \left\{ -iO_a(x), -iO_b(y) \dots \right\} \Psi_\alpha^+ \right), \quad (6.4.3)$$

where $O_a(x)$, etc. are the counterparts of $o_a(x)$ in the Heisenberg picture

$$O_a(\mathbf{x}, t) = \exp(iHt) o_a(\mathbf{x}, 0) \exp(-iHt) = \Omega(t) o_a(\mathbf{x}, 0) \Omega^{-1}(t), \quad (6.4.4)$$

$$\Omega(t) \equiv e^{iHt} e^{-iH_0 t}, \quad (6.4.5)$$

and Ψ_β^+ and Ψ_β^- are 'in' and 'out' eigenstates of the full Hamiltonian H , respectively.

Here is the proof. From Eq. (3.5.10), we see immediately that the left-hand side of Eq. (6.4.3) is

$$\begin{aligned} \left[\frac{\delta^r S[\epsilon]}{\delta\epsilon_{a_1}(x_1) \dots \delta\epsilon_{a_r}(x_r)} \right]_{\epsilon=0} &= \sum_{N=0}^{\infty} \frac{(-i)^{N+r}}{N!} \int_{-\infty}^{\infty} d\tau_1 \dots d\tau_N \\ &\times \left(\Phi_\beta, T \left\{ V(\tau_1) \dots V(\tau_N) o_{a_1}(x_1) \dots o_{a_r}(x_r) \right\} \Phi_\alpha \right). \end{aligned} \quad (6.4.6)$$

For definiteness, suppose that $x_1^0 \geq x_2^0 \geq \dots \geq x_r^0$. Then we can denote by $\tau_{01} \dots \tau_{0N_0}$ all τ s greater than x_1^0 ; by $\tau_{11} \dots \tau_{1N_1}$ all τ s between x_1^0 and x_2^0 , and so on; finally denoting by $\tau_{r1} \dots \tau_{rN_r}$ all τ s that are less than x_r^0 .

* For a single O operator, this is a version of the Schwinger action principle.⁴